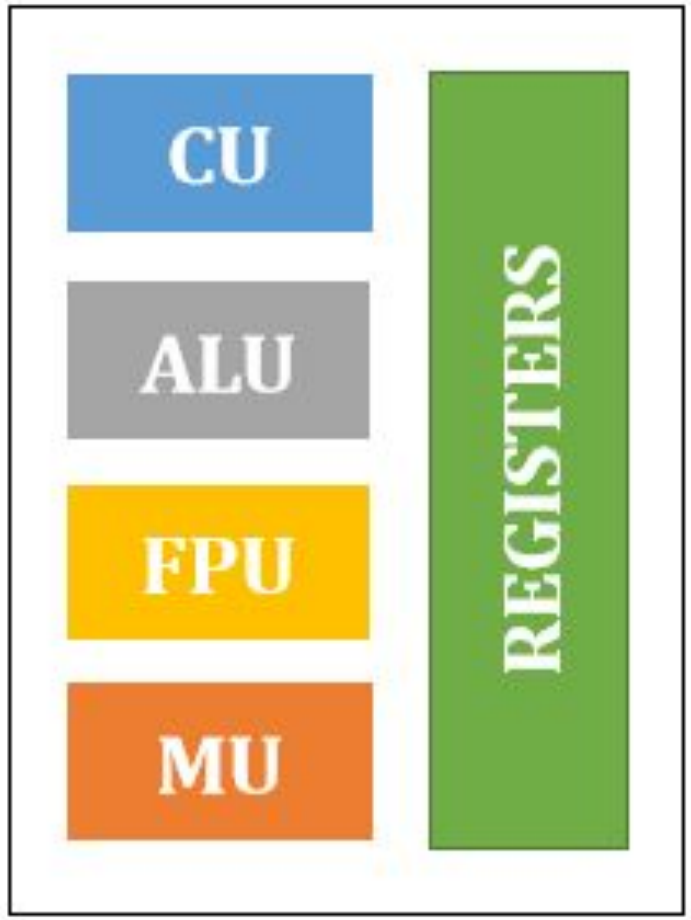
Prerequisite - Introduction to parts of CPU



- Arithmetic and Logic Unit (ALU)
  It performs all the arithmetic and logical micro operations.
- Floating Point Unit (FPU)
  It performs operations on floating point numbers.
- Memory Unit (MU)
  It stores the set of instructions.
- Control Unit (CU)
  It supervises the sequence of micro operations.
- Registers
  Temporary storage area, which holds the data during the execution of an instruction.

Prerequisite - Registers inside CPU



Addressing Modes

- Microprocessor executes the instructions stored in memory (RAM).
- It executes one instruction at a time.
- Each of the instruction contains operations and operands.
- Operation specifies the type of action to be performed.
  - For example: ADD, SUB, MOV, INC, LOAD, STORE
- Operands are the data on which the operation is to be performed.

MOV B, A Here MOV is operation and (B & A) are operands.

ADD B Here ADD is operation and (B) is operand.

Addressing Modes

- Operand can be place either in one of the processor **register** or in **memory**.
- There are different ways to get the operands.
- The way in which the operand is taken from register or memory is named as **addressing mode**.

Addressing Modes

1. Immediate Addressing Mode
2. Register Addressing Mode
3. Register Indirect Addressing Mode
4. Direct Addressing Mode
5. Indirect Addressing Mode
6. Implied Addressing Mode
7. Relative Addressing Mode
8. Indexed Addressing Mode
9. Base Register Addressing Mode
10. Autoincrement or Autodecrement Addressing Mode

1. Immediate Addressing Mode

- The operand is specified with in the instruction.
- Operand itself is provided in the instruction rather than its address.

Move Immediate

MVI A , 15h $A \leftarrow 15h$ Here 15h is the immediate operand

Add Immediate

ADI 3Eh $A \leftarrow A + 3Eh$ Here 3Eh is the immediate operand

2. Register Addressing Mode

- The operand is specified with in one of the processor register.
- Instruction specifies the register in which the operand is stored.

Move

MOV C , A $C \leftarrow A$ Here A is the operand specified in register

Add

ADD B $A \leftarrow A + B$ Here B is the operand specified in register

3. Register Indirect Addressing Mode

- The instruction specifies the register in which the memory address of operand is placed.
- It do not specify the operand itself but its location with in the memory where operand is placed.

Move

MOV A , M $A \leftarrow [[H][L]]$

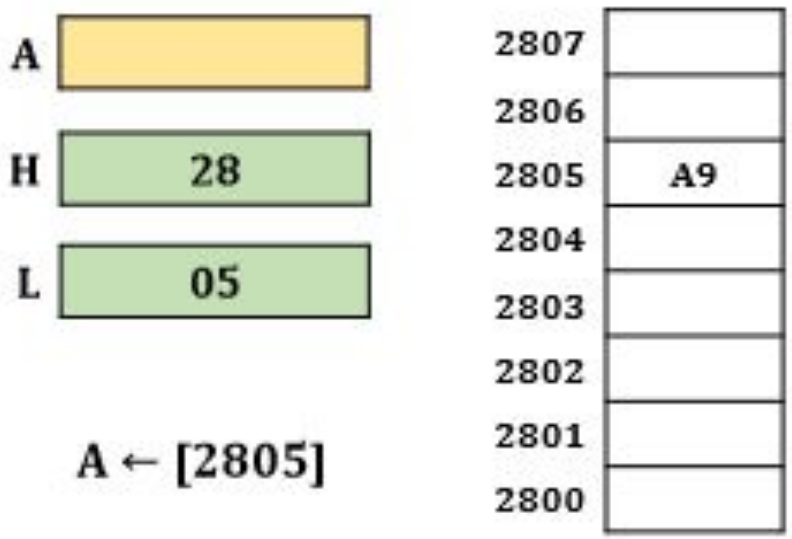
It moves the data from memory location specified by HL register pair to A.

3. Register Indirect Addressing Mode

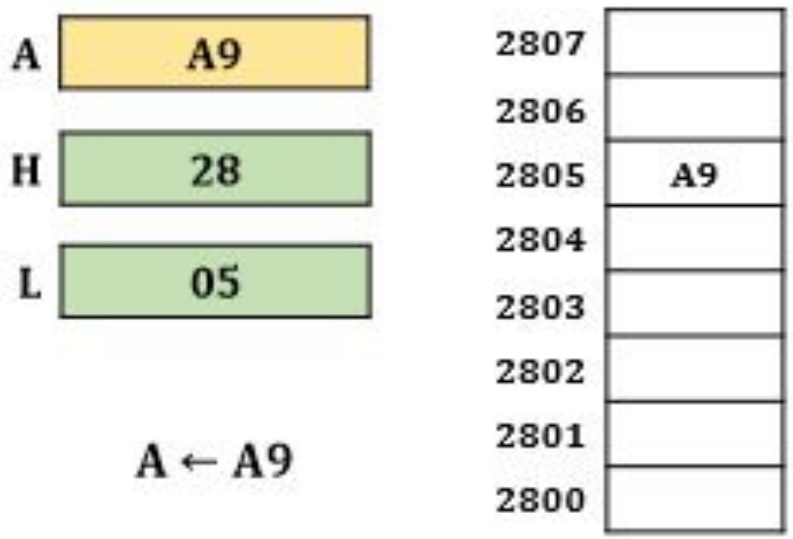
MOV A , M $A \leftarrow [[H][L]]$

It moves the data from memory location specified by HL register pair to A.

Before



After



4. Direct Addressing Mode

- The instruction specifies the direct address of the operand.
- The memory address is specified where the actual operand is.

Load Accumulator

LDA 2805h $A \leftarrow [2805]$

It loads the data from memory location 2805 to A.

Store Accumulator

STA 2803h $[2803] \leftarrow A$

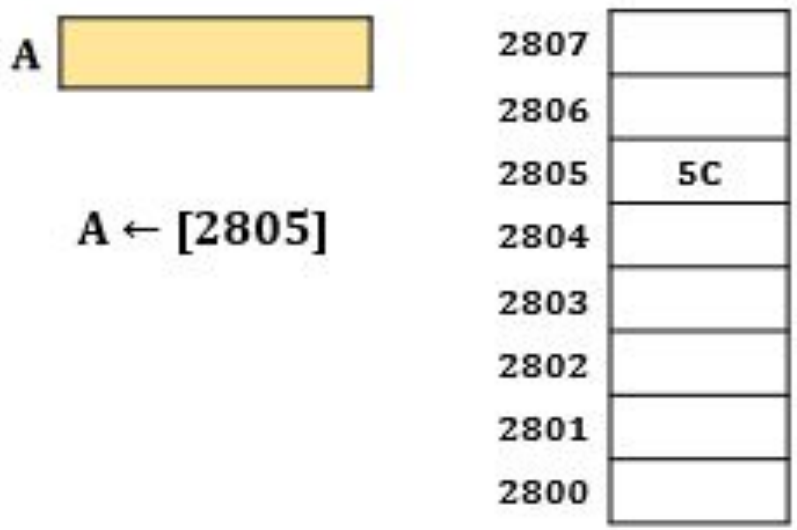
It stores the data from A to memory location 2803.

4. Direct Addressing Mode

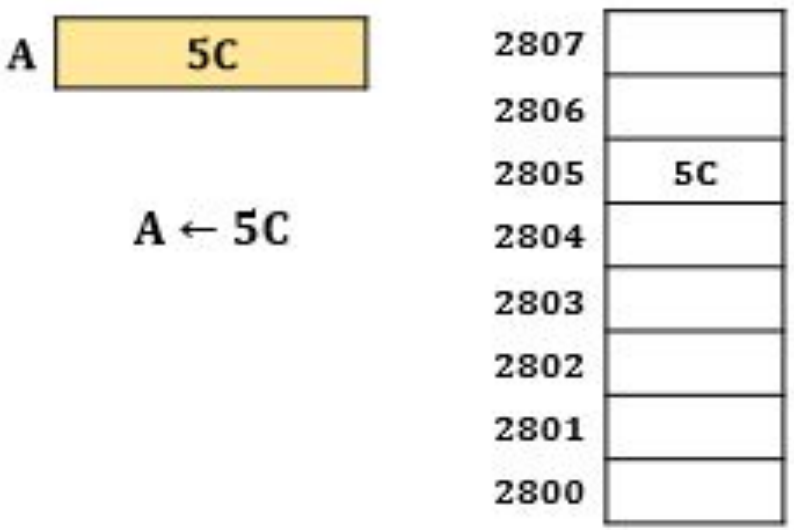
LDA 2805h A ← [2805]

It loads the data from memory location 2805 to A.

Before



After

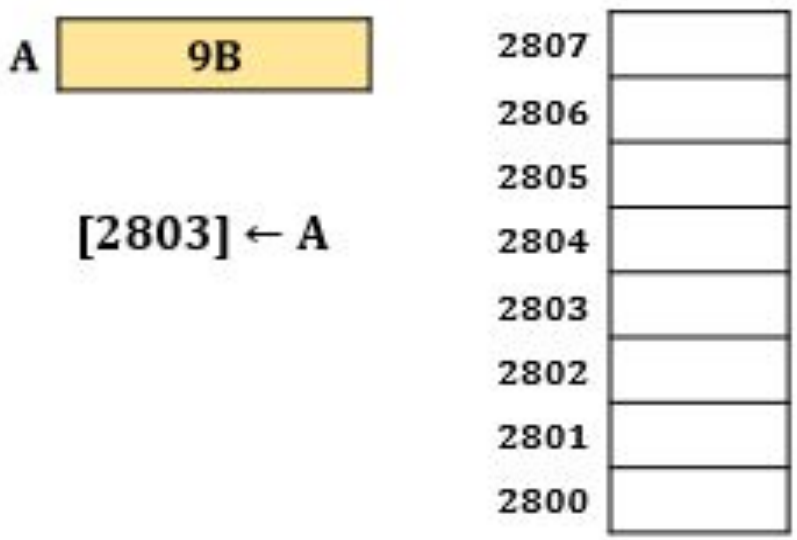


4. Direct Addressing Mode

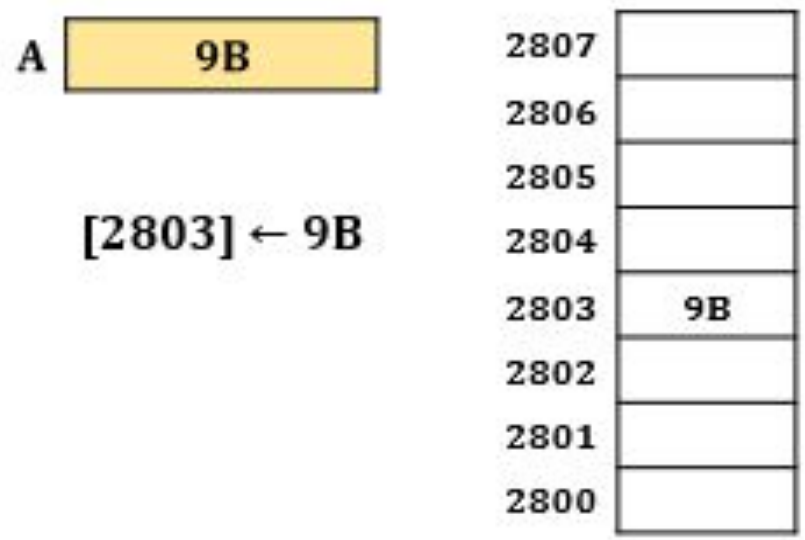
STA 2803h [2803] ← A

It stores the data from A to memory location 2803.

Before



After



5. Indirect Addressing Mode

- The instruction specifies the indirect address where the effective address of the operand is placed.
- The memory address is specified where the actual address of operand is placed.

Move

MOV A, 2802h A ← [[2802]]

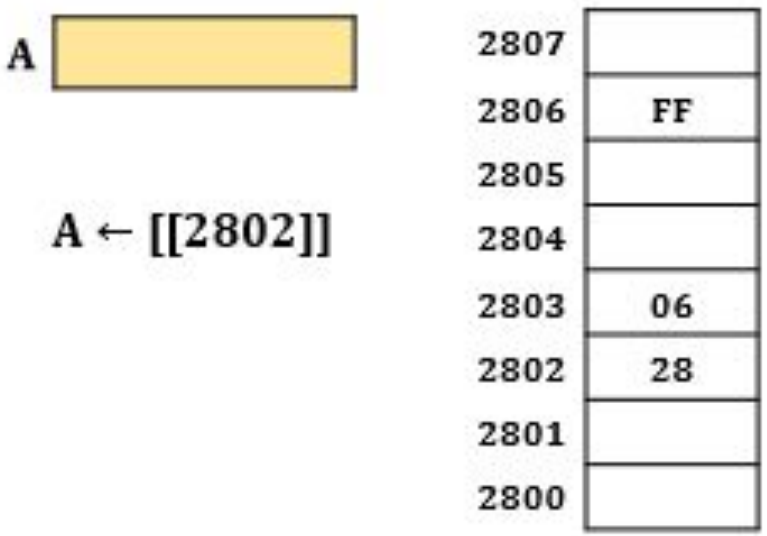
It moves the data from memory location specified by the location 2802 to A.

5. Indirect Addressing Mode

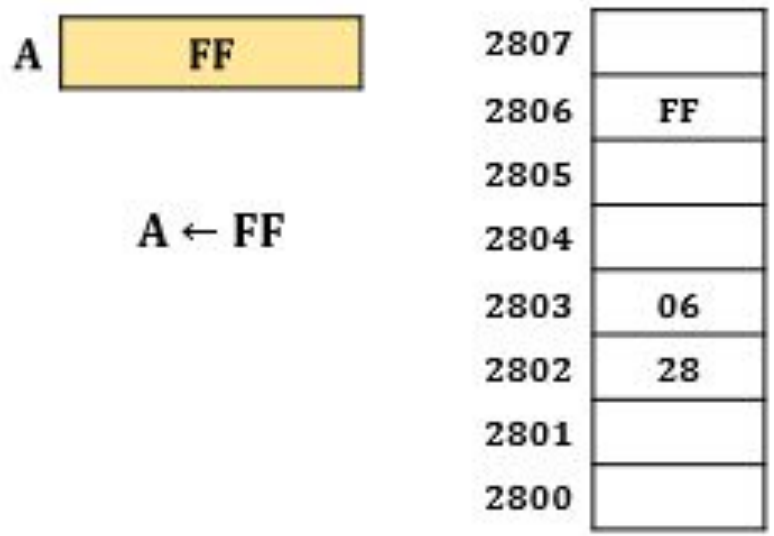
MOV A, 2802h A ← [[2802]]

It moves the data from memory location specified by the location 2802 to A.

Before



After



6. Implied Addressing Mode

- It is also called inherent addressing mode.
- The operand is implied by the instruction.
- The operand is hidden/fixed inside the instruction.

Complement Accumulator **CMA**
(Here accumulator A is implied by the instruction)

Complement Carry Flag **CMC**
(Here Flags register is implied by the instruction)

Set Carry Flag **STC**
(Here Flags register is implied by the instruction)

7. Relative Addressing Mode

- In relative addressing mode, contents of Program Counter PC is added to address part of instruction to obtain effective address.

- The address part of the instruction is called as offset and it can +ve or –ve.

- When the offset is added to the PC the resultant number is the memory location where the operand will be placed.

7. Relative Addressing Mode

Offset = 04h

PC | **2801**

2807	22
2806	FF
2805	6D
2804	59
2803	08
2802	2E
2801	F3
2800	9F

2807	22
2806	FF
2805	6D
2804	59
2803	08
2802	2E
2801	F3
2800	9F

Actual Operand

Effective address of operand = **PC + 01 + offset**
Effective address of operand = **2801 + 01 + 04**
Effective address of operand = **2806h**

7. Relative Addressing Mode

Offset = 03h

PC | **2803**

2807	22
2806	FF
2805	6D
2804	59
2803	08
2802	2E
2801	F3
2800	9F

2807	22	**Actual Operand**
2806	FF	
2805	6D	
2804	59	
2803	08	
2802	2E	
2801	F3	
2800	9F	

Effective address of operand = **PC + 01 + offset**
Effective address of operand = **2803 + 01 + 03**
Effective address of operand = **2807h**

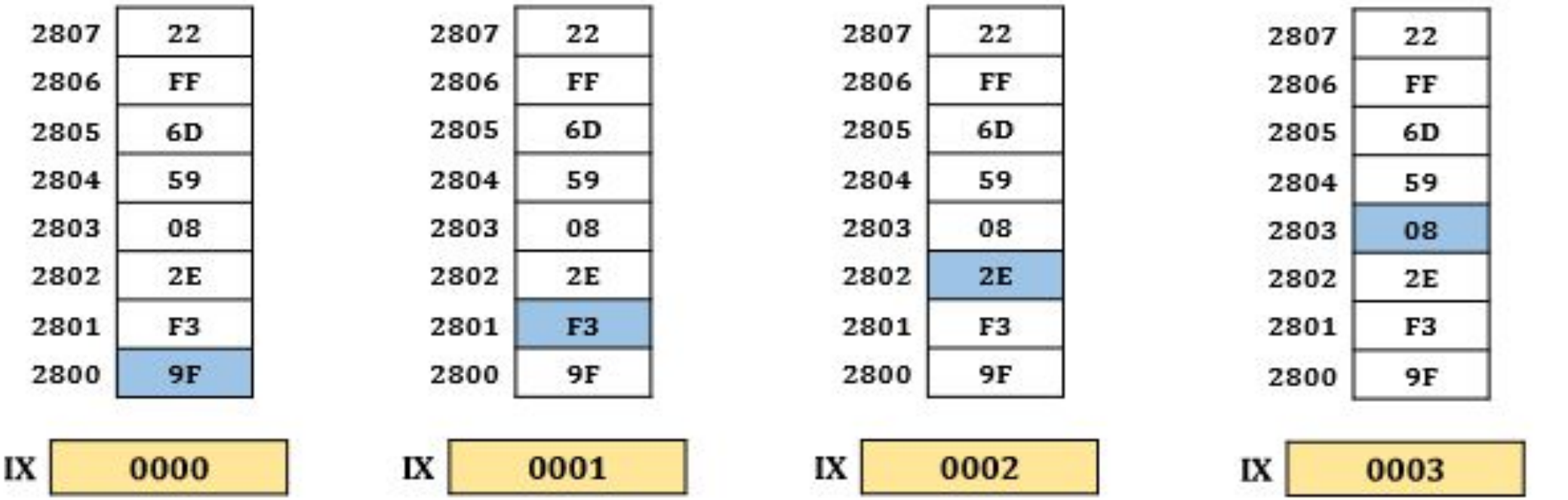
8. Indexed Addressing Mode

- In index addressing mode, contents of Index register is added to address part of instruction to obtain effective address.
- The address part of instruction holds the beginning/base address and is called as base.
- The index register hold the index value, which is +ve.
- Base remains same, the index changes.
- When the base is added to the index register the resultant number is the memory location where the operand will be placed.

8. Indexed Addressing Mode

Base = 2800h

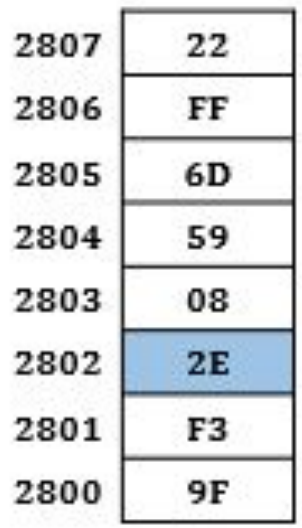
Effective address of operand = **Base + IX**



8. Indexed Addressing Mode

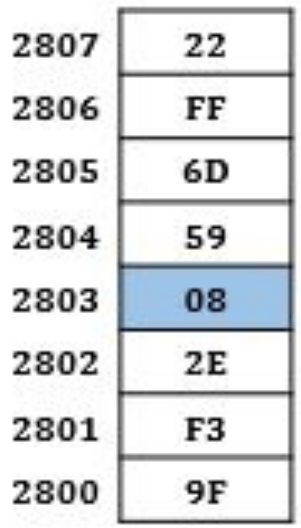
Base = 2802h

Effective address of operand = **Base + IX**



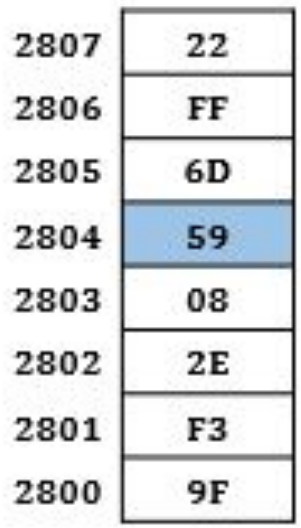
IX 0000

2802h + 0000h = 2802h



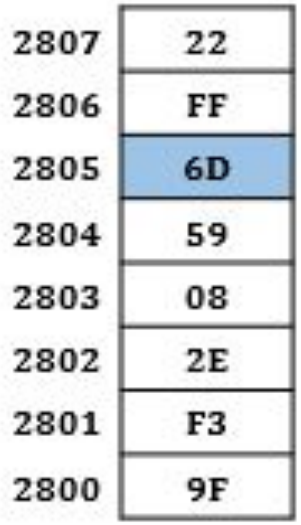
IX 0001

2802h + 0001h = 2803h



IX 0002

2802h + 0002h = 2804h



IX 0003

2802h + 0003h = 2805h

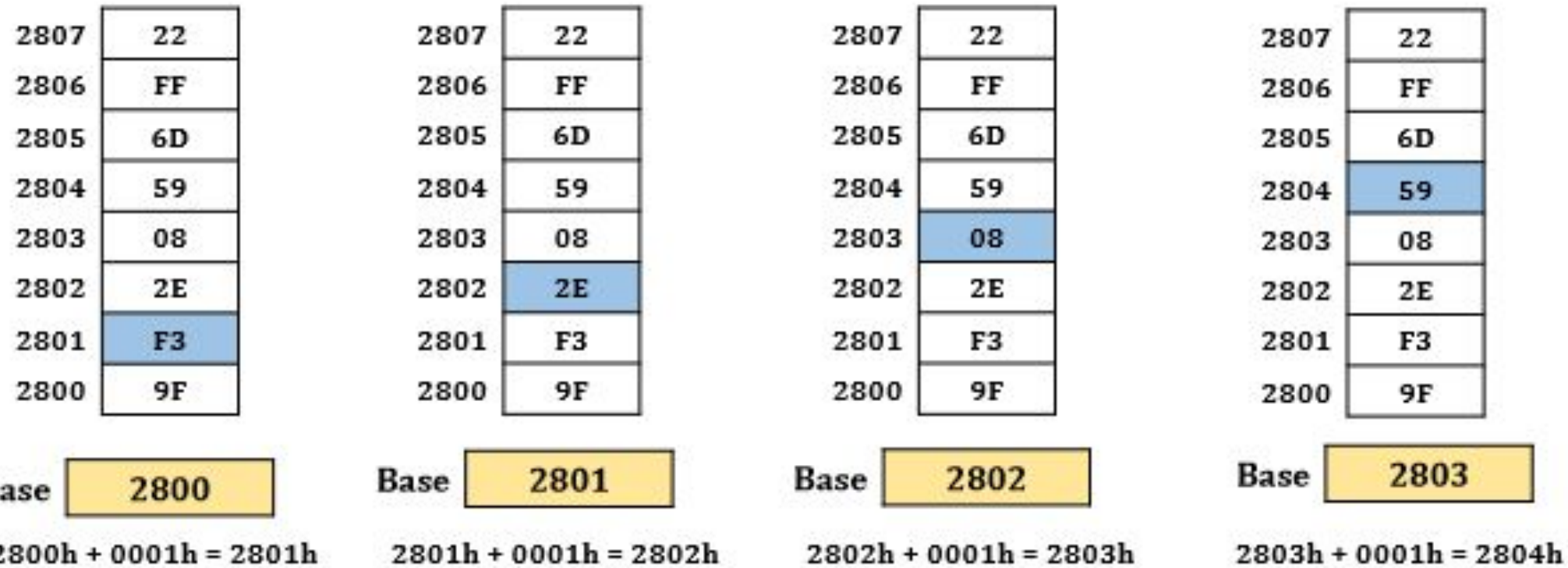
9. Base Register Addressing Mode

- In base register addressing mode, contents of base register is added to address part of instruction to obtain effective address.
- It is similar to the indexed addressing mode except the register now is called as base instead of index.
- The base register hold the beginning/base address.
- The address part of instruction holds the offset.
- Offset remains same, the base changes.
- When the offset is added to the base register the resultant number is the memory location where the operand will be placed.

9. Base Register Addressing Mode

Offset= 0001h

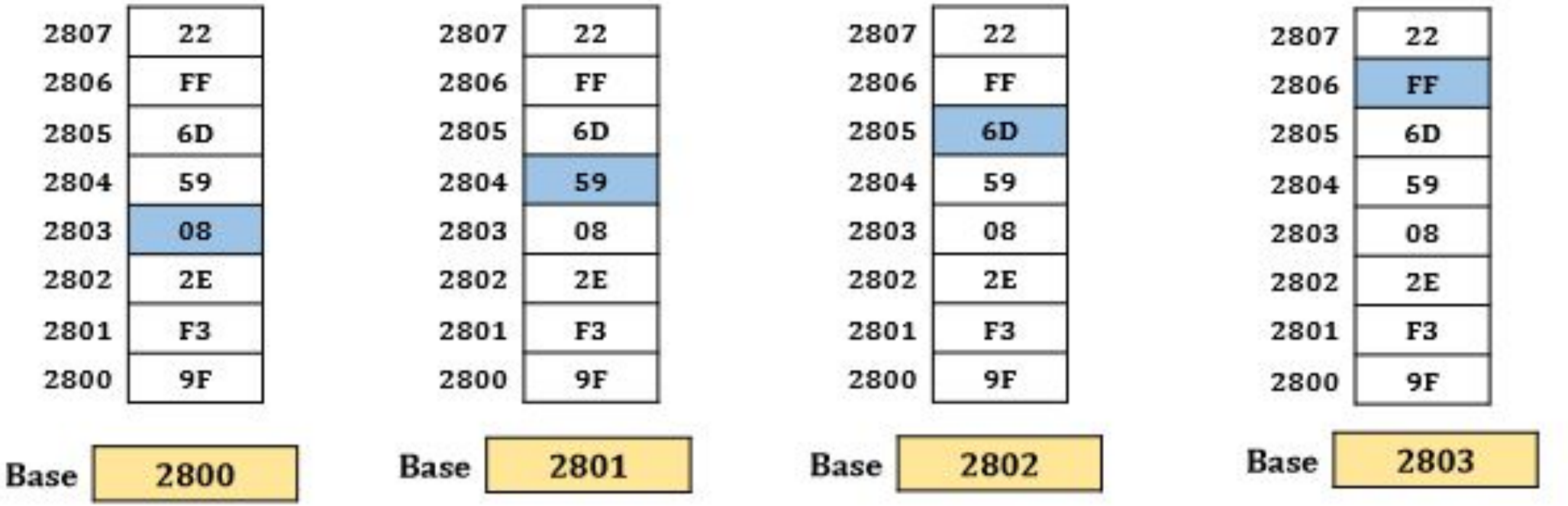
Effective address of operand = **Base Register + offset**



9. Base Register Addressing Mode

Offset= 0003h

Effective address of operand = **Base Register + offset**



10. Autoincrement or Autodecrement Addressing Mode

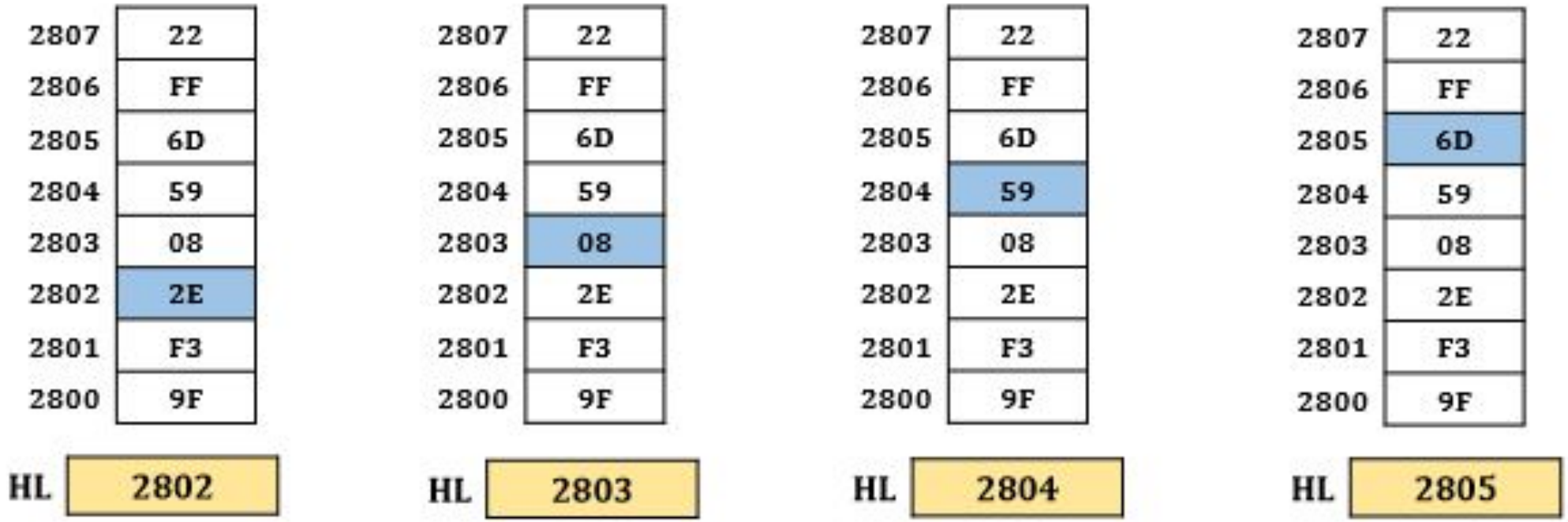
- It is similar to register indirect addressing mode.

- Here the register is incremented or decremented before or after its value is used.

10. Autoincrement or Autodecrement Addressing Mode

HL pair incremented after its value is used

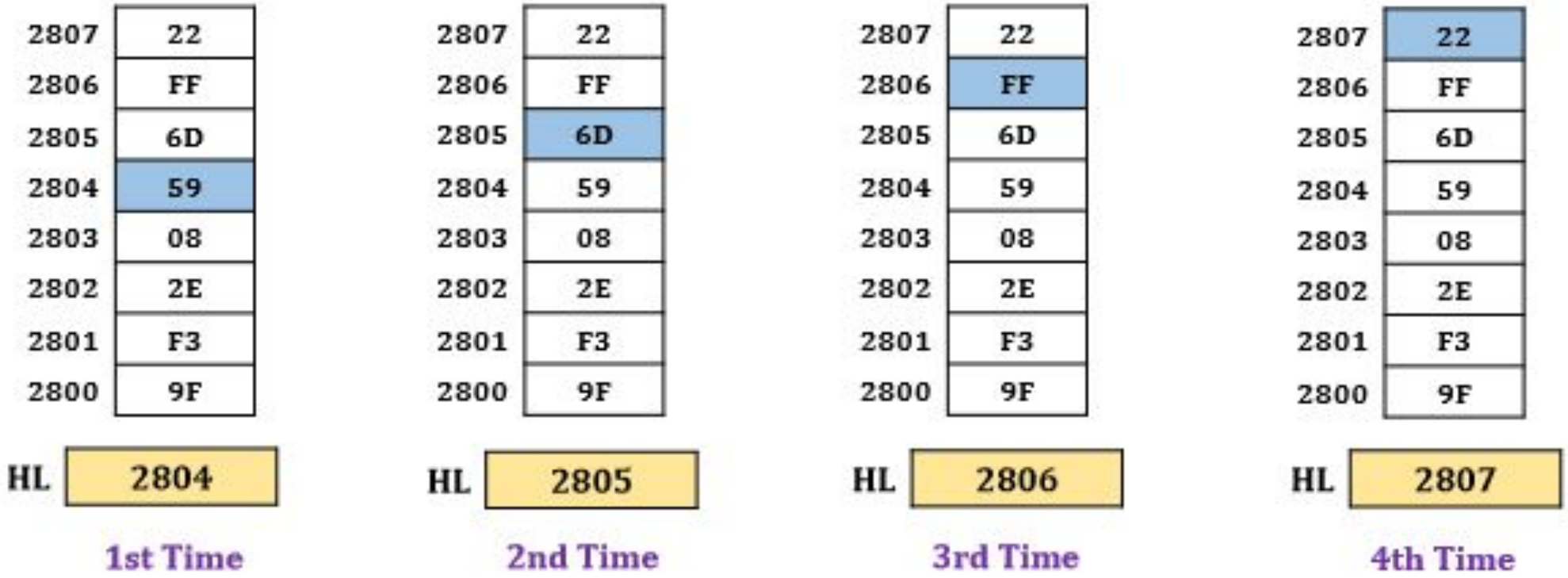
At start: HL 2802



10. Autoincrement or Autodecrement Addressing Mode

HL pair incremented before its value is used

At start: HL 2803



Example problem

PC	200
R1	400
XR	100
AC	

Address	Memory	
200	Load to AC	Mode
201	Address = 500	
202	Next Instruction	
399	450	
400	700	
500	800	
600	900	
702	325	
800	300	

PC = Program Counter
R1 = Register
XR = Index Register
AC = Accumulator

- Memory is having first instruction to load AC
- Mode will specify the addressing mode to get operand.
- Address field of instruction is 500.

Find out the effective address of operand and operand value by considering different addressing modes.

Example problem

PC	200
R1	400
XR	100
AC	

Address	Memory	
200	Load to AC	Mode
201	Address = 500	
202	Next Instruction	
399	450	
400	700	
500	800	
600	900	
702	325	
800	300	

1. Immediate Addressing Mode

- As instruction contains immediate number 500.
- It is stored as address 201.

Effective Address = 201
Operand = 500

AC	500

Example problem

PC	200
R1	400
XR	100
AC	

Address	Memory	
200	Load to AC	Mode
201	Address = 500	
202	Next Instruction	
399	450	
400	700	
500	800	
600	900	
702	325	
800	300	

2. Register Addressing Mode

- Register R1 contains 400.
- As operand is in register so no any memory location.

Effective Address = Nil
Operand = 400

AC	400

Example problem

PC	200
R1	400
XR	100
AC	

Address	Memory	
200	Load to AC	Mode
201	Address = 500	
202	Next Instruction	
399	450	
400	700	
500	800	
600	900	
702	325	
800	300	

3. Register Indirect Addressing Mode

- Register R1 contains 400.
- So effective address of operand is 400.
- The data stored at 400 is 700.

Effective Address = 400
Operand = 700

AC	700

Example problem

PC	200
R1	400
XR	100
AC	

Address	Memory	
200	Load to AC	Mode
201	Address = 500	
202	Next Instruction	
399	450	
400	700	
500	800	
600	900	
702	325	
800	300	

4. Direct Addressing Mode

- Instruction contains the address 500.
- So effective address of operand is 500.
- The data stored at 500 is 800.

Effective Address = 500
Operand = 800

AC	800

Example problem

PC	200
R1	400
XR	100
AC	

Address	Memory	
200	Load to AC	Mode
201	Address = 500	
202	Next Instruction	
399	450	
400	700	
500	800	
600	900	
702	325	
800	300	

5. Indirect Addressing Mode

- Instruction contains the address 500.
- Address at 500 is 800.
- So effective address of operand is 800.
- The data stored at 800 is 300.

Effective Address = 800
Operand = 300

AC	300

Example problem

PC	200
R1	400
XR	100
AC	

Address	Memory	
200	Load to AC	Mode
201	Address = 500	
202	Next Instruction	
399	450	
400	700	
500	800	
600	900	
702	325	
800	300	

6. Relative Addressing Mode

- PC = 200.
- Offset = 500.
- Instruction is of 2 bytes.
- So effective address = PC + 2 + offset = 200 + 500 +2 = 702 .
- The data stored at 702 is 325.

Effective Address = 702

Operand = 325

AC	325

Example problem

PC	200
R1	400
XR	100
AC	

Address	Memory	
200	Load to AC	Mode
201	Address = 500	
202	Next Instruction	
399	450	
400	700	
500	800	
600	900	
702	325	
800	300	

7. Index Addressing Mode

- XR = 100.
- Base = 500.
- So effective address = Base + XR = 500 + 100 = 600 .
- The data stored at 600 is 900.

Effective Address = 600
Operand = 900

AC	900

Example problem

PC	200
R1	400
XR	100
AC	

Address	Memory	
200	Load to AC	Mode
201	Address = 500	
202	Next Instruction	
399	450	
400	700	
500	800	
600	900	
702	325	
800	300	

8. Autoincrement Addressing Mode

- It is same as register indirect addressing mode except the contents of R1 are incremented after the execution.
- R1 contains 400.
- So effective address of operand is 400.
- The data stored at 400 is 700.

Effective Address = 400
Operand = 700

R1	401
AC	700

Example problem

PC	200
R1	400
XR	100
AC	

Address	Memory	
200	Load to AC	Mode
201	Address = 500	
202	Next Instruction	
399	450	
400	700	
500	800	
600	900	
702	325	
800	300	

9. Autodecrement Addressing Mode

- It is same as register indirect addressing mode except the contents of R1 are decremented before the execution.
- R1 contains 400.
- R1 is first decremented to 399.
- So effective address of operand is 399.
- The data stored at 399 is 450.

Effective Address = 399
Operand = 450

R1	399
AC	450

Example problem

Addressing Mode	Effective Address	Operand
Immediate Addressing Mode	201	500
Register Addressing Mode	Nil	400
Register Indirect Addressing Mode	400	700
Direct Addressing Mode	500	800
Indirect Addressing Mode	800	300
Relative Addressing Mode	702	325
Indexed Addressing Mode	600	900
Autoincrement Addressing Mode	400	700
Autodecrement Addressing Mode	399	450